OpenSky Global Flight Analysis

Aviation Data Analytics & Flight Behavior Insights

Akash Kumar Gautam
TANISHK AGRAWAL
Kshtri Yatin Singh
Navodit Sharma
Ravleen Singh
Ritik

Context & Problem Statement

Context:
Modern aviation generates large volumes of aircraft state data through ADS-B systems. However, this data is often unstructured and noisy, making analysis difficult.

Core Problem:
Raw OpenSky data cannot be directly used to understand flight behavior, aircraft distribution, or operational patterns.

Objective:
To transform raw aviation data into a structured dataset and extract meaningful insights for analysis.

Data Engineering (Source to Pipeline)

Source Dataset:

- OpenSky Network API (Aircraft State Vectors)
- ~8,000–10,000 records (single snapshot)

Python ETL Pipeline:

- Removed missing latitude/longitude values
- Handled missing altitude and velocity
- Standardized data types and formats

Feature Engineering:

- Created time-based fields
- Transformed heading using sin/cos
- Prepared final dataset for analysis

KPI & Metrics Framework

Data Overview KPIs

- Total Aircraft Records → Overall dataset size
- Airborne Records → Number of aircraft in flight
- Grounded Records → Aircraft on ground

Operational KPIs

- Airborne Ratio → Airborne aircraft / total records
- Average Altitude → Typical flight level of aircraft
- Average Velocity → Speed behavior of aircraft

Distribution KPIs

- Country-wise Aircraft Distribution → Regional spread of aircraft
- Velocity Distribution → Spread of aircraft speeds
- Altitude Distribution → Spread of flight levels

Data Quality KPIs

- Missing Values Handled → Clean dataset after preprocessing
- Outliers Identified → Extreme values flagged using IQR
- Inconsistent Records Flagged → e.g., low velocity in airborne state

Key Insights (EDA)

- Aircraft distribution is uneven across regions
- Majority of aircraft are observed in airborne state
- Most aircraft operate within standard cruise altitude ranges
- Lower altitude cluster represents takeoff/landing phases
- Some records show inconsistencies in velocity values

Advanced Analysis

Outlier Detection:

- Applied IQR method on velocity and altitude

Segmentation:

- Group-based processing for better data consistency

Feature Transformation:

- Converted heading into sin/cos components
- Helps handle circular directional data

Tableau Dashboard Walkthrough

Executive View:

- Global aircraft density map
- KPI summary (aircraft, altitude, velocity)
- Top countries by aircraft count

Operational View:

- Velocity vs altitude scatter plot
- Flight phase distribution (climbing, descending, level)
- Velocity distribution histogram

Recommendations

Improve Data Quality

- Apply validation rules for inconsistent records

Expand Dataset

- Collect multiple snapshots for time-based analysis

Improve Data Validation

- Apply stricter checks on altitude, velocity, and flight status to reduce inconsistent records

Enhance Feature Engineering

- Create additional derived variables for deeper insights

Impact & Value

- Improved data reliability by handling missing and inconsistent records
- Better understanding of aircraft behavior through structured analysis

- Clean and structured dataset ready for analysis and visualization
- More accurate insights by reducing noise in the data

- Easier pattern interpretation using dashboard visualizations
- Strong foundation for future analytics and extensions

LIMITATIONS & NEXT STEPS

Limitation	Next Step
• Single snapshot dataset (no time-based analysis) • Geographic bias due to uneven ADS-B coverage • No flight-level tracking across time • Limited contextual data (e.g., weather, routes)	• Collect time-series data for trend analysis • Develop advanced analytical or predictive models • Integrate external datasets (weather, routes) • Build real-time or frequently updated dashboard

OpenSky Global Flight Analysis

Thank You